

Case 1: New Skill Acquisition [alloca_strat_005]

Coverage Expansion for Second-Order Cone Optimization Problems

Source Problem: OptMATH #145

This is a **Second-Order Cone Programming (SOCP)** problem. Minimize a linear objective over $x[0], x[1], x[2]$, subject to three equality constraints ...

Retrieval Skill: No matching strategy and experience found

Execution Results: misused addGenConstrNorm for SOC constraint

Failure Diagnosis: miss strategy → has no SOC-structured modeling strategy

New Strategy: alloca_strat_005

Summary: Model a linear objective with linear equalities and SOC constraints of the form $\sum(y^2) \leq t^2 \dots$

Procedure: Define primary and auxiliary variables; add objective, equality/linking constraints, and SOC constraints...

New Problem: OptMATH #149

Minimize a linear cost over $x[0], \dots, x[4]$, subject to four equality constraints...

Retrieval Skill: (new) alloca_strat_005

Execution Results: get correct answer -5.07

Case 2: Repair Skill Update [route_exp_002]

Boundary Refinement for MTZ Subtour Elimination in TSP

Source Problem: MamoComplex #84

This is a **Traveling Salesman Problem (TSP)**. A delivery driver must visit 5 shops once and return to the starting shop, minimizing total travel cost...

Retrieval Skill: route_strat_001 + route_exp_002

Execution Results: model becomes infeasible

Failure Diagnosis: misleading experience → fixed start position $u_start = 0$

Repaired Experience: route_exp_002

Trigger: Use MTZ subtour elimination for TSP with auxiliary position variables.

Guidance: Fix the starting node's position to 0 and bound all other nodes

between 1 and $n-1$; [Add] apply MTZ constraints only to pairs where both nodes are **not the starting node**.

New Problem: OptMATH #66

A logistics company must deliver packages to 7 cities, visit each city exactly once, and form a single continuous loop with minimum total delivery cost...

Retrieval Skill: route_strat_001 + (repaired) route_exp_002 + route_exp_005

Execution Results: get correct answer 1998